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MILITARY SPECIFICATION

INTRAVENOUS INJECTION SET, VENTED, WITH METERING CHAMBER WITHOUT NEEDLE, DISPOSABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of sterile, disposable, vented intravenous injection set with metering chamber suitable for accurate administration of parenteral solutions in controlled volumes from 10 ± 2 ml to 100 ml at 60 drops per ml from glass containers without airways.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-RST, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-601	- Rubber: Sampling and Testing.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection. by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-147	- Palletized Unit Loads.
MIL-STD-794	- Part and Equipment, Procedures for Packaging and Packing of.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/HIMA MD70.1 Luer Fittings for Medical Materiel.

(Application for copies should be addressed to the Publication Sales, Dept STD, American Institute of Physics, 335 E. 45th St., NY, NY 10017.)

U. S. PHARMACOPEIAL CONVENTION, INC.
Pharmacopeia of the United States (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE

Federal Food, Drug, and Cosmetic Act and
General Regulations for its enforcement.

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(Application for copies should be addressed to the Food and Drug Administration, U.S. Department of Health, Education, and Welfare, Washington, DC 20204.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material.

3.1.1 Rubber and plastic components. All rubber and plastic components shall be of the best quality and suitable in all respects for the intended purpose. All rubber and plastic components shall be sterile, free from pyrogens, compatible, and safe (nontoxic) when tested as specified in Section 4. Those rubber and plastic components coming in contact with solutions being administered, and end protectors, shall contain no more than 5 parts per million (p.p.m.) total of lead, barium and cadmium when tested as specified in 4.4.1. The plastics shall meet the requirements of the Pharmacopeia of the United States (USP) for Class IV plastics and for Transfusion and Infusion Assemblies, when tested as specified in 4.4.11. Each lot of product need not be biologically tested provided evidence is furnished that the formulation had previously been tested, that, it complies with USP requirements for Class IV plastics, and that it is the same as being offered on the contract.

3.1.1.1 Molded rubber parts. The molded rubber parts shall be fabricated from natural rubber.

3.1.1.2 Plastic tubing. Plastic tubing shall be nontoxic, transparent, colorless polyvinyl chloride. The tubing shall be flexible and pliable, but sufficiently rigid to prevent collapse during normal use. The tubing shall be free from all liquid substances which oxidize, exude, or sweat out to the surface of the tubing.

3.1.1.3 Piercing device. The piercing device shall be white and fabricated from a suitable rigid plastic.

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3.1.1.4 Needle adapter. The needle adapter shall be fabricated from a suitable rigid plastic.

3.1.1.5 End protectors. End protectors shall be fabricated from suitable plastic sheaths or caps.

3.1.1.6 Medicinal entry assembly (or Y injection site). The medicinal entry assembly shall be assembled from a part molded of a suitable plastic, a rubber diaphragm or plug stopper and a stopper securing band.

3.1.1.6.1 Rubber diaphragm or plug. The rubber diaphragm or plug on the medicinal entry assembly leg shall be molded from natural, non-oxidizing, low-sulfur, wax-free rubber 45 ± 5 Shore A Durometer.

3.1.1.7 Metering chamber. The metering chamber shall be fabricated of transparent cellulose propionate 0.022 inch plus .003 minus .007 inch wall thickness in a cylindrical shape forming a semi-rigid chamber. The plastic shall be free from all liquid substances which oxidize, exude or sweat out to the surface of the film.

3.1.1.8 Flow control clamp. The roller-type control clamp shall be fabricated from a suitable rigid plastic.

3.1.2 Slide clamp. The slide clamp shall be fabricated from suitable rigid plastic.

3.1.3 Other material. All other materials hereinafter specified shall be of the best quality and suitable in all respects for the purpose intended.

3.2 Style and design. Style and design shall be in accordance with Figure 1 and as hereinafter specified.

3.2.1 Description. The set shall be a sterile, non-pyrogenic, non-toxic disposable intravenous injection set for use in accurate administration of parenteral solutions from standard solution containers (see 3.2.1.2). The set shall consist of a piercing device, a drip chamber, a shut-off clamp, an in-line graduated metering chamber, a medicinal entry assembly, a length of plastic tubing, a Y or flash bulb injection site, one flow control clamp, an air vent with bacterial filter, a needle adapter, and end protectors. A slide clamp or valve shall be present on the air inlet to enable shutting off the air flow. The piercing device shall be provided with a convenient hand grip for the insertion of the piercing device into the stopper or port of the solution container. The assembly of the set shall be structurally sound and shall not leak air or fluid during use.

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Open ends of the set shall be covered (vented as required for passage of sterilizing gases) with readily removable plastic covers, or end protectors, which shall maintain the sterility and freedom from pyrogens of the fluid path after sterilization.

3.2.1.1 Piercing device. The piercing device shall consist of an integrally molded unit including a spike and an air inlet. The piercing device shall be a dual-channeled, conical spike, one channel of which shall serve as an air inlet. The spike shall be 7/8 to 1 inch in length, with a diameter of 0.258 inch plus 0.008 inch, minus 0.008 at the base. The spike tip shall be sharpened to be suitable for puncture of and insertion into the stopper or port of a standard solution container (hereinafter specified) without undue pressure (see 3.2.1.2.1). Edges of fluid and air inlet passage openings shall be round or dull to prevent stopper coring. The spike shall be ruggedly designed to prevent coring or shearing, and shall be structurally sound. It shall be firmly retained in a leakproof manner when inserted in the stopper or port of a standard solution container. The piercing device shall not break or leak air or fluid during use. It shall not be brittle or soft, or become brittle or soft under normal storage conditions. The standard solution container stopper referenced above shall be in accordance with requirements contained in the current specification for NSN 6505-00-975-0604, Dextrose Injection, USP. The passages in the piercing device shall be properly designed to prevent air from passing directly from the air passage to the fluid passage under maximum flow conditions. The air inlet passage shall vent the solution container (without the need of an airway cannula) during administration of solution. This air inlet opening shall have a female Luer taper which shall be capped with a removable and replaceable air filter to prevent the ingress of bacteria (see 3.2.1.2.2). The filter cap shall form a snug fit with, and shall be retained on, the opening until manually removed. The fluid passage in the spike shall extend into the connected plastic input tubing.

3.2.1.2 Durability of piercing device. When tested as specified in 4.4.3, the device shall not loosen or permit leakage, nor shall the device become damaged. The piercing device shall be capable of being inserted into the standard solution containers, as described in 4.4.3, without difficulty, the tip (or point) of the device not breaking or bending to the degree of causing difficulty in insertion into standard solution containers.

3.2.1.2.1 Air inletting bacterial filter. The air inletting bacterial filter shall be of such design and material as to comply with the following performance requirements when tested as specified in 4.4.4:

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(a) Bacterial retention. The filter shall be such that the rate of contamination of the test media shall be not greater than 0.1% (1-1000).

(b) Air passage. The filter shall pass air at a rate exceeding 150 ml per minute when tested in accordance with 4.4.4.2.

(c) Resistance to water. The filter shall support a 30-inch static head of water for 45 seconds with no wetting of the filter element.

3.2.1.2 Plastic input tubing. The entry plastic tubing which connects the piercing device assembly to the metering chamber shall be a 5 to 7 inch length of transparent plastic tubing with an inside diameter of 0.120 to 0.125 and a wall thickness of 0.025 to 0.032 inch. The plastic tubing shall be connected to the piercing device or drip chamber by expansion and the plastic tubing shall be connected on the opposite end to the entry end of the metering chamber. The attachments shall be tight and leakproof when tested in accordance with 4.4.9. A captive slide clamp designed to close off solution flow shall be positioned on this tubing.

3.2.1.3 Slide clamp. The slide clamp shall be a rectangular piece of plastic nominally 1-5/8 inch long with a longitudinal opening tapered and slotted to provide an open and closed position. The slot (opening) shall be of such dimensions that when the tubing is at the wider end of the taper (open position) the clamp is capable of being rotated 360° around the tube without binding or scoring the tube, and shall allow unrestricted solution flow. When the tubing is forced into the narrow end of the slot (closed position), fluid flow shall be completely shut-off. Exterior corners and edges of the clamp shall be slightly beveled to prevent the clamp from cutting the fingers during the application of pressure required to open or close the clamp. Edges and sides of the slot shall be slightly beveled and smooth enough to prevent the clamp from cutting the tube when in either the open or closed position. The action of opening and closing the clamp shall not cut the tubing when tested in accordance with 4.4.13.

3.2.1.4 Medicinal entry assembly. The upper medicinal entry assembly shall be an integral part of the entry end of the metering chamber. It shall consist of a rubber diaphgram and securing band, and shall provide for direct injection into the metering chamber.

The lower medicinal entry assembly shall be of the Y-type design fabricated of a clear, suitable rigid plastic with two legs in line and the third leg at a nominal 30° from the other. Plastic tubing shall be permanently attached in a leakproof manner to two legs. The third leg, which is in line with one leg and at nominal 30° of the other, shall be fitted with a rubber diaphragm and securing band. There may be, in addition, a flash bulb located between the outlet tubing and the needle adapter.

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3.2.1.5 Rubber diaphragm. The rubber diaphragm, plug or flash bulb for medicinal entry shall be molded from natural, non-oxidizing, low-sulfur, wax-free rubber of 45 ± 5 or, as an alternate 33 ± 5 Shore A Durometer hardness. The diaphragm or plug shall be so secured that after puncture by and withdrawal of the needle in accordance with 4.4.12 there shall be no separation or leakage.

3.2.1.6 Metering chamber. For Figure 1, the metering chamber shall be constructed of material as specified in 3.1.1.7. The overall length of the tubing forming the 100 ml chamber, shall be a nominal 5 to 6 inches with an internal diameter of about 1.25 to 1.40 inches. The housing shall be scaled from 0 to 100 ml. or to as much as 150 ml in 1 ml. calibrations with numerical designations at least every 10 ml. These calibrations shall be permanently imprinted on the housing. The 150 ml chamber shall be proportionally larger and graduated to 150 ml.

The upper end of the housing shall be closed, in a tight leakproof manner, by a rigid plastic lid that has integrally molded into its upper surface, projections with openings. The opening positioned at the center of the lid receives the tubing connected to a fitment on the piercing device. The other openings are spaced equidistant around the center opening. One opening shall be closed by a gum rubber plug that is held securely with a plastic band insuring security during puncture by and withdrawal of a needle. The gum rubber insert shall reseal after puncture with a dry 18 gauge needle in accordance with 4.4.12. Permanently attached to the remaining opening is a piece of medical grade polyvinyl chloride tubing that passes through the throat of a slide clamp and terminates in a female Luer adapter which is covered by a removable air inlet passage device of polyolefin incorporating a bacterial filter media made of hydrophobic material. The bottom of the metering chamber shall be closed by a soft elastomeric valve or membrane constructed so that upon depletion of the solution there shall be a positive shut-off of the solution flow.

3.2.1.7 Pediatric drip chamber. The drip chamber shall be constructed of a flexible, transparent plastic as specified in 3.1.1 to permit squeezing by hand and yet sturdy enough to resist collapse during normal manipulation. The drip chamber shall have an effective volume between 3 ml. minimum and 7 ml. maximum. The fluid passage in the chamber shall be calibrated to deliver 60 ± 8 drops per ml. when tested as specified in 4.4.2 and be designed to eliminate the formation of macro-drops (larger than the desired size); the housing shall be of an extended length to provide assurance against air entrapment in the set in the event of high flow rate delivery.

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3.2.1.8 Plastic outlet tubing. The plastic outlet tubing shall be fabricated of material as specified in 3.1.1.2 and shall have an inside diameter of 0.100 to 0.105 and a nominal wall thickness 0.019 and shall be 54 inches ± 2 inches in length. The overall length of the set is a nominal 75 inches +9, -3 inches. The fittings shall be firmly and permanently secured at both ends to provide air tight, leakproof connection and shall not separate when tested as specified in 4.4.9.

3.2.1.9 Flash bulb. The flash blub shall be a minimum of 1-1/2 inches and a maximum of 2-1/4 inches in length, and shall be constructed with an enlarged center section approximately 5/8 inch in length, and a minimum wall thickness of 0.060 inch. The center section shall have a shoulder on one end to provide for easy insertion of an 18 gage needle when adding supplemental medication. The flash bulb shall be capable of resealing subsequent to puncture with a nonsiliconized 18 gage needle cannula. When tested for resealing as specified in 4.4.12 there shall be no falling drop within 10 seconds.

3.2.1.10 Control clamp. Control clamp shall be fabricated of rigid plastic, shall be a roller-clamp from which the roller cannot be accidentally removed, and shall be supplied in an open position. The clamp shall provide a constant setting and quick shut-off from the operating position, shall be movable, and shall be capable of one-handed operation. To assure constant delivery of fluid, it shall be impossible for the clamp setting to be changed drastically by an accidental jerk on the tubing. When tested as specified in 4.4.5, the clamp shall be capable of accomplishing complete quick shut-off (i.e. no falling drop in drip chamber or from needle adapter (3.2.1.11), shall regulate flow of the solution from complete shut-off to maximum flow, and with a constant head of water shall maintain a constant flow rate at any point between complete shut-off and maximum flow. In addition, when the clamp is operated from its fully open position to complete shut-off, the tubing shall not be cut, pierced, or damaged so as to cause leakage or restriction of flow of solution when tested in accordance with 4.4.5. The roller shall not be removed from the roller clamp body when subjected to a 3.65 kilogram load in accordance with 4.4.5.1.

3.2.1.11 Needle adapter. Needle adapter shall be fabricated of suitable plastic and so constructed as to be attached at one end to the rubber flash bulb, if present, or directly to the tubing in a leakproof manner. The other end of the adapter shall be tapered to fit a needle having a standard Luer hub in accordance with ANSI/HIMA Standard MD70.1 in a tight leakproof manner (see 3.7). The adapter, or the assembly of the adapter to the flash bulb, shall be such as to prevent accidental reversal of the assembly of the adapter to the rubber component at time of assembly. The needle adapter shall not separate from the flash bulb or tubing when tested as specified in 4.4.9.

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3.2.1.12 End protectors. End protectors shall be readily removable sheaths or molded caps constructed in such a manner that when placed over each open end of the intravenous injection set they shall, after sterilization, maintain sterility, freedom from pyrogens, and integrity of the entire fluid path of the infusion device until manually removed. Absorbent cotton shall not be present in the end protectors.

3.3 Manufacturer's record. The manufacturer shall place a lot control number on the paper label, or on the printed carton, which can be verified against the manufacturer's records of all tests made available for inspection by any duly qualified medical officer of the Armed Forces, National Institutes of Health, any official of the Federal Food and Drug Administration, or any duly authorized Government representative. No shipment shall be made until all the tests have been completed, verified, and recorded in the permanent records of the manufacturer.

3.4 Sterility, safety, and freedom from pyrogens. All equipment shall be safe (non-toxic). Fluid paths shall be sterile and pyrogen-free. The fluid path shall be pyrogen-free when tested as specified in 4.4.6 and sterile when tested as specified in 4.4.7. The sets shall be safe when tested for safety as specified in 4.4.8.

3.5 Certificates. Certificates of sterility, freedom from pyrogens, safety, and bacterial retention as specified below, shall be submitted as follows for destination inspection. Certificates shall be submitted by the contractor directly to Technical & Quality Assurance Division, Directorate of Medical Materiel, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, PA 19101, Attention: Quality Assurance Branch, and one copy of each shall be forwarded with each shipment to the consignee; for source inspection, certificates shall be furnished to the procuring activity via the government inspector.

3.5.1 Certificates of sterility. A certificate of sterility, stating that samples from each sterilizer lot have been tested and found to be sterile, shall be submitted to the procuring activity. The certificate of sterility shall include:

a. Item identification.
b. National Stock Number.
c. Contract or purchase order number.
d. Sterility lot or control number.
e. Date of sterilization.
f. Quantity of product in sterilizer lot.
g. Type of sterilization (gas, radiation, heat, steam, etc.).

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h. Form of biological indicator (i.e., inoculated paper spore strip or carrier, inoculated product, or inoculated simulated product).
i. Test organism(s) and population density used.
j. Test medium, incubation time, and dates of tests.
k. Number of test units.
l. Statement that samples comply with sterility tests.
m. Signature of head of laboratory or a designated qualified alternate.

3.5.1.1 Sterility lot. A sterility lot is that single quantity of product subjected to the same manufacturing operation and simultaneously sterilized in the same sterilizing chamber. Each unit package shall be clearly marked with a sterility lot (control) number as specified in 5.3.1. Each sterility lot shall be traceable to the pertinent manufacturing lot(s).

3.5.2 Certificate of freedom from pyrogens. A certificate of freedom from pyrogens, listing all lot or control numbers, shall state that samples from each lot or control have been tested for pyrogens as specified in 3.4 and were found to comply with such requirements. The certificate shall reference the method used in determining freedom from pyrogens.

3.5.3 Certificate of safety. A certificate of safety, listing all lot or control numbers, shall state that samples from each lot or control have been tested for safety as specified in 3.4 and were found to comply with such requirements. The certificate shall reference the method used in determining safety.

3.6 Compatibility. The materials of which the sets are manufactured shall not react with, or be acted upon by, the liquids with which they will be in contact during normal use and operation, nor shall there be any materials incorporated in the set which will cause any deleterious effect when the set is used for the purpose intended.

3.7 Leakage and separation of set. Sets shall be tested for leakage and separation of set as specified in 4.4.9. Sets which show leakage or separation at any part of the set shall be rejected.

3.8 Solutions flow rate. When tested as specified in 4.4.10, the intravenous injection set, with the filter in its normal dry condition, shall permit the passage of 100 cc. of fluid in not more than 5-1/2 minutes, and in not more than 8-1/2 minutes after the filter has been immersed in water in accordance with 4.4.10.

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3.9 Instructions. A detailed instruction sheet for proper use of the sets, single and in tandem, shall be provided in each unit package.

3.10 Workmanship. Workmanship shall be first class throughout. The intravenous injection sets shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical menas of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the plastic, metal and rubber, may be furnished by lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number (NSN), the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in Table I. The unit of product for examination shall be one intravenous injection set.

TABLE I. Sampling for examination

Type of defect	Inspection level	AQL (percent defective)
Major	S-4	1.0
Minor	S-4	2.5

4.2.2 For tests.

4.2.2.1 For sterility test. Sampling for sterility test shall be in accordance with the directions in 4.4.7.

4.2.2.2 For pyrogen, and safety tests. Sampling procedures for pyrogen, and safety tests shall be conducted as indicated in Table II.

4.2.2.3 For heavy metals. Samples shall be taken from each lot of rubber or plastic as indicated in Table II.

4.2.3 For other tests. Sampling for other tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table II.

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TABLE II. Sampling for other tests

Characteristic	Requirement	Test procedure	Insp. level	AQL (percent defective)
Class IV Plastics	3.1.1	4.4.11	S-1	*
Heavy metals	3.1.1	4.4.1	S-1	*
Air Filter	3.2.1.2.2	4.4.4		
		4.4.4.2	S-2	*
		4.4.4.3	S-2	*
		4.4.4.1	4.4.4.1	*
Piercing device durability	3.2.1.2.1	4.4.3	S-3	*
Slide clamp	3.2.1.4	4.4.13	S-3	*
Sterility	3.4	4.4.7	see4.4.7	*
Pyrogens	3.4	4.4.6	S-2	*
Safety	3.4	4.4.8	S-2	*
Resealing	3.2.1.5.1 and 3.2.1.9	4.4.12	S-3	*
Drip chamber	3.2.1.7	4.4.2	S-3	*
Control clamp	3.2.1.10	4.4.5 and 4.4.5.1	S-3	*
Needle adapter	3.2.1.11	4.4.9	S-3	*
Leakage and separation of set	3.7	4.4.9	S-3	*
Solution flow	3.8	4.4.10	S-3	*

*Acceptance number shall be zero for all sample sizes.

4.2.4 Sample unit usage. Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.2.5 For bacterial retention test. This test shall be performed at least once for each contract. The samples selected shall represent the exact composition, treatment, and construction to be used throughout the entire contract. Any change in the filter shall require additional testing. The number of samples specified in 4.4.4.1 shall be randomly selected from production lots of filters used. Failure of any sample to comply shall be cause for rejection of the filters represented.

4.2.6 For other tests. Sample size for each test shall be as indicated for inspection levels of MIL-STD-105 as shown in Table II. The acceptance number shall be zero for all sample sizes.

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4.3 Examination. The sets shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted in accordance with MIL-STD-105 and the classification of defects specified in Table III. Inspection, however, shall not be restricted to the classified possible defects listed in Table III.

TABLE III. Classification of defects

Categories	Defects
Critical	None defined.
Major	
101	End protectors missing or dislodged.
102	Protector not free of holes.
103	Tubing and metering chamber not free of bloom or tackiness.
104	Set incomplete.
105	Style and design not as specified.
106	Presence of foreign matter.
107	Dimensions not as specified.
Minor	
201	Instructions missing, incomplete, illegible.
202	Contractor's name or registered trademark, missing, incomplete, or illegible.

4.4. Tests.

4.4.1 Heavy metals extractable test. A 2-gram composite sample shall be obtained from the rubber and plastic components. The material shall be cut into small silvers, so that no thickness exceeds 3 mm., and eluted for 72 hours at 50°C., 24 hours 70°C., or 60 minutes at 121°C. in one menstrum; 0.9% sodium chloride. Each gram of the material shall be covered with 5 milliliters (ml.) of the menstruum. The menstruua are collected, agitated after elution, and then submitted to spectographic atomic absorption spectrophotometric or specific wet analysis for lead, barium, and cadmiun. The metals are reported in parts per million. Thus 1.0 ppm in elute is equivalent to extracting 1.0 mg. from 200 g of test material dispersed in 1,000 ml. of elution menstruum.

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4.4.1.1 Alternate heavy metals extractable test. Accurately weigh a 5-gram sample for each of the metals (barium, cadmium or lead) into a 1-oz. Vycor crucible. Reduce to ash in a muffle furnace at 650°C for eight hours. Let cool to room temperature and add 15 ml. of 1:1 hydrochloric acid. Bring to a boil on a hot plate, allow to boil for 2 to 3 minutes, remove, let cool, and filter through Whatman #2 filter paper into a 50-ml. volumetric flask. Rinse the residue on the filter with deionized water and dilute to 50 ml. Prepare standard solutions of 1, 2, and 5 ppm for each of the metals, namely barium, cadmium, and lead. Determine the absorbances of these solutions at appropriate wave lengths using a hollow Cathode Ba, Cd, or Pb lamp for each of the respective metals and a 10X scale expansion on a spectrophotometer, such as Perkin-Elmer Model 303, atomic Absorption Spectrophotometer. Determine the absorbance of the sample solutions using the same conditions as with the standards.

Calculations:

A = Absorbance
F = Factor = $\frac{\text{(Concentration standard in ppm Ba, Cd, or Pb)}}{\text{absorbance}}$*

W = Sample weight in grams
*The three standards are averaged to give ppm/unit A.

4.4.2 Drop test. The intravenous injection set shall be tested at room temperature, using water at a rate of 100 ± 5 drops per minute. The number of drops required to deliver 2 ml. of water shall be counted, and the drops per ml. calculated from this value.

4.4.3 Piercing device durability test. Samples of finished sets, after sterilization, shall be used for this test. The piercing (on assembled set) shall be inserted into, and removed from, a standard solution container (specified in 3.2.1.2) in the normal manner. This operation shall be repeated twice. The piercing device shall then be immersed in a solution of 5 percent dextrose in isotonic sodium chloride solution (in accordance with requirements of the USP monograph for "Dextrose and Sodium Chloride Injections") for 4 hours. The piercing device shall then be removed from the solution and immediately inserted into a standard solution container as specified in 3.2.1.2. Clamp base of spike in a suitable fixture, apply a five-pound load perpendicular to longitudinal axis at 1/8" from sharpened tip, load to be applied for a period of one minute. While the spike is still in the fixture apply a five-pound load perpendicular to longitudinal axis at a point 1/16" from end of tubing connector for a period of one minute. There shall be no fracture as a result of this test.

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4.4.4 Air inletting bacterial filter.

4.4.4.1 Bacterial retention. The following test shall be performed on not less than 50 sample filters lotted to represent specific materials used in production batches of filters, and four additional test bottles shall be used as controls for each test period: Detach drip chamber assembly from each sample sterile intravenous injection set by sealing and cutting the plastic tube within 2 inches of the drip chamber so that the tubing interior remains sterile. Regular parenteral 1000 ml. (1-liter) bottles having a standard closure shall be filled with 50 ml of fluid thioglycollate medium, USP, evacuated to at least 25 inches of mercury (Hg), sealed and sterilized. Introduce a sufficient concentration of A. aerogenes into a No. 44 De Vilbiss pocket nebulizer, or similar apparatus. Connect nebulizer to air filter on a drip chamber assembly by means of gum rubber tubing with necessary reducers and connectors. Adjust the concentration of organisms in the nebulizer to obtain a minimum of 300 organisms in the air volume replacing the vacuum in the liter test bottle. Insert a piercing device of each prepared sample drip chamber into a bottle closure in the normal manner. When the vacuum in the liter test bottle has been displaced, disconnect the testing apparatus from the filter and shake the bottle thoroughly to mix the aerosol content and to wet all surfaces. Retain the drip chamber assembly in the bottle for venting and incubate at 31° ± 1°C. for seven days. As a control to assure that the aerosol is challenging the filter correctly, remove the air filter from the drip chamber assembly of two additional bottles at the start and two additional bottles at the end of each and every test period and connect the nebulizer. When the vacuum has been displayed, disconnect the nebulizer and replace the filter. After mixing the bottle contents, determine the bacterial count of the bottle. The rate of contamination shall be no greater than specified in 3.2.1.2.2.

4.4.4.2 Air passage. The air-inletting bacterial filter shall pass air as specified in 3.2.1.2.2, when tested assembled in the filter cap and under a pressure equal to a column of 16 inches of water. The flow shall be measured with a Brooks Rotameter, or equivalent apparatus.

4.4.4.3 Resistance to water. The air-inletting bacterial filter shall be tested for resistance to water in its filter cap by placing the filter cap on a fitting connected to a 30-inch column of water. The filter shall be examined for wetting at the end of 45 seconds exposure.

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4.4.5 Clamping device performance test. The assembled set shall be connected in the normal manner, to a standard solution container as specified in 3.2.1.2 except, as an alternate, the container may be filled with saline solution or water. The container, with set connected thereto, shall be suspended, in the normal manner, at a sufficient height to permit the entire length of the tubing of the set to hang vertically. The ability of the clamp to accomplish complete shut off, to regulate flow, and with a constant head of water maintain a constant rate of flow, shall be tested with the clamp at a point on the plastic tubing within two inches above the flash bulb or Y Injection Site. The tests shall be performed with the tubing hanging vertically at all times. The rate of flow shall be determined either by counting the drops in the drip chamber during a measured time period, or by recording the time required to collect a measured volume of fluid from the needle adapter. The control clamp shall be operated from its fully open position to complete shut off and this operation repeated 10 times while the clamp is in the same position on the tubing and shall meet the requirements specified in 3.2.1.10.

4.4.5.1 Security of roller in clamp body. The clamp with the specified tubing in place and extending one inch beyond each end of the clamp shall be held vertically, with the large end down, over a supporting gap so that the roller if forced out of the clamp body is free to fall. A rod of steel is applied tangentially to the top of the roller. The rod area making contact with the roller shall be flat and polished. The vertical downward force required to separate the roller from the clamp body is measured. As an alternate, any position of the clamp is acceptable providing the clamp body is not distorted and the load measurement is accurate.

4.4.6 Freedom from pyrogens. Samples of the finished sets shall be tested for freedom from pyrogens, in accordance with the USP, for Transfusion and Infusion assemblies.

4.4.7 Sterility test.

The intravenous injection set shall be tested for sterility in accordance with the method specified in the U.S.P. using inoculated product, inoculated simulated product, or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates it from product intended for distribution. The incubation medium for the paper spore strip (carrier) shall be soybean-casein digest. The incubation media for the inoculated product, or inoculated simulated product shall be fluid thioglycollate or alternative thioglycollate at 30° to 35°C for one half of the specimens and soybean-casein digest at 20° to 25°C for the remaining half of the specimens. The minimum incubation time shall be ten days. Sampling shall be as follows:

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Medium	Paper Spore Strip	Inoc.Prod. or Inoc. Simulated Prod.
Soybean-casein digest	S-3*	S-3*
Fluid thioglycollate or Alternative thioglycollate		S-3*

*Use a minimum of 20 specimens. Acceptance number is zero.

4.4.7.1 Inoculated product or simulated product. For each type of sterilization, the organism used shall be as listed below:

Type of Sterilization	Organism
Steam	B. stearothermophilus
Dry heat	B. stearothermophilus
	B. subtilis
Ethylene oxide	B. subtilis
Gamma radiation	B. pumilus

The number of viable spores shall be sufficient to provide assurance that the product has been sterilized.

4.4.7.2 Inoculated carrier. Inoculated carriers (paper spore strips) if used, shall be in accordance with MIL-S-36586A, Sterilization Test Strip Set, Bacterial Spore. Other paper carriers shall be inoculated to be of equivalent resistance. The method of incubation and number of test units shall be as indicated in 4.4.7. Sterility test and 4.4.7.1. Inoculated product or simulated product. The method of sterilization will control the organism in the spore strip to be tested for as shown in 4.4.7.1. NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the U.S.P. and paragraphs 4.4.7 and 4.4.7.1.

4.4.7.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Requests for approval of any alternate method of testing for sterility shall be made to the Contracting Officer, and approval obtained before employment of the alternate method on a government contract.

4.4.7.4 Product testing. Testing for sterility of the product after receipt by the government shall be without biological indicator and in accordance with the USP. The inspection level shall be as stated in paragraph 4.4.7.

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4.4.8 Safety (toxicity) test. The finished sets shall be tested for safety in accordance with the USP for Transfusion and Infusion Assemblies.

4.4.9 Leakage and separation of set. This test shall be applied to those samples subjected to the piercing device durability test (see 4.4.3). The set shall be subjected to an internal hydrostatic pressure of 20 pounds per square inch (psig). No falling drops shall be noted in the 3 minutes following application of the pressure.

4.4.10 Solution flow rate. The finished intravenous injection set shall be connected to a standard solution container as specified in 3.2.1.2 in the normal manner, the entire assembly inverted and suspended so that there is a 30 inch column of fluid in the set and container and no restriction of flow imposed by the clamp. The time necessary for passage of 100 cc. of fluid through the entire set shall be determined. This test procedure shall be repeated after purposely immersing the filter with water for a period of at least 10 minutes.

4.4.11 Biological test. The plastics shall be tested in accordance with the USP for Class IV plastics.

4.4.12 Reseal test. The rubber injection sites shall reseal without a falling drop within 10 seconds after puncture with a nonlubricated 18 gage needle when subjected to an internal hydrostatic pressure of 4 p.s.i.g. for 30 seconds. The needle shall be thoroughly wiped with a dry, clean cloth several times to remove any lubricant that may be present before insertion.

4.4.13 Slide clamp test. Each slide clamp shall be operated for 10 complete cycles from stop flow to full flow with the clamp located approximately in center of tubing length while subjected to an internal hydrostatic pressure of 4 p.s.i.g. At the completion of the test the tubing shall be examined for cuts.

4.5 Inspection of packaging. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A.

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5.1.1.1 Unit package. Each set shall be packaged in a box constructed in accordance with PPP-B-566, variety, style, type and class optional. The box shall be designed with interior fittings to prevent the tubing from being kinked or taking a permanent set. The box shall be further wrapped with polypropylene film. The film shall be seated in such a manner so as to prevent undetected tampering or access to the contents unless damaged or opened.

5.1.1.2 Intermediate package. Twenty-four unit packages shall be packaged in an intermediate package constructed in accordance with PPP-B-636, class domestic. Closure of the box shall be as specified in the box specification.

5.1.2 Level C. The level C preservation for intravenous injection sets shall conform to the MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Two intermediate packages (48 units) shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Intravenous injection sets shall be packed as specified in 5.2.1 except that the exterior shall be class domestic and strapping and waterproofing shall not be required.

5.2.3 Level C. The level C packing for intravenous injection sets shall conform to the MIL-STD-794 requirements for this level.

5.2.4 Unitized loads. Unitized loads commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination exceed 250 pounds or 20 cubic feet. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 20 cubic feet need not be unitized. Pallet loads including the pallet, shall not exceed 54 inches in height, 48 inches in width and 40 inches in length. Pallet shall conform to Type IV or Type V.

5.2.4.1 Level A & B. Intravenous injection sets packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets in conformance with MIL-STD-147.

5.2.4.2 Level C. Intravenous injection set, packed as specified in 5.2.3, shall be unitized as specified in MIL-STD-794.

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5.3 Marking.

5.3.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129 and the requirements of the Federal Food, Drug and Cosmetic Act. Marking shall also include the lot (control) number and name of contractor. The legend: "FLUID PATH AND AREAS COVERED BY END PROTECTORS ARE STERILE AND PYROGEN FREE" or similar statement shall also be shown.

5.3.2 Level A, B and C. Each intermediate package, exterior container and unitized load shall be marked as specified in MIL-STD-129. The lot (control) number shall also be shown.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

6.1 Intended use. The intravenous injection set is intended for accurate administration of parenteral solutions in controlled volume from 10 ± 2 ml to 100 ml at 60 drops per ml from plastic or glass containers without airways.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquistion documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) National Stock Number.

6.3 National Stock Number (NSN) coverage. This specification covers the following National Stock Number:

National Stock No.	Item Identification
6515-00-433-2089	INTRAVENOUS INJECTION SET, Vented, With Metering Chamber, Without Needle, Disposable

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6.4 Subject term (key word) listiing.

Injection set
Intravenous injection set

MILITARY INTERESTS:

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing Activity:
DoD - MB
Agent
DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQA

PROJECT NO. 6515-38.1

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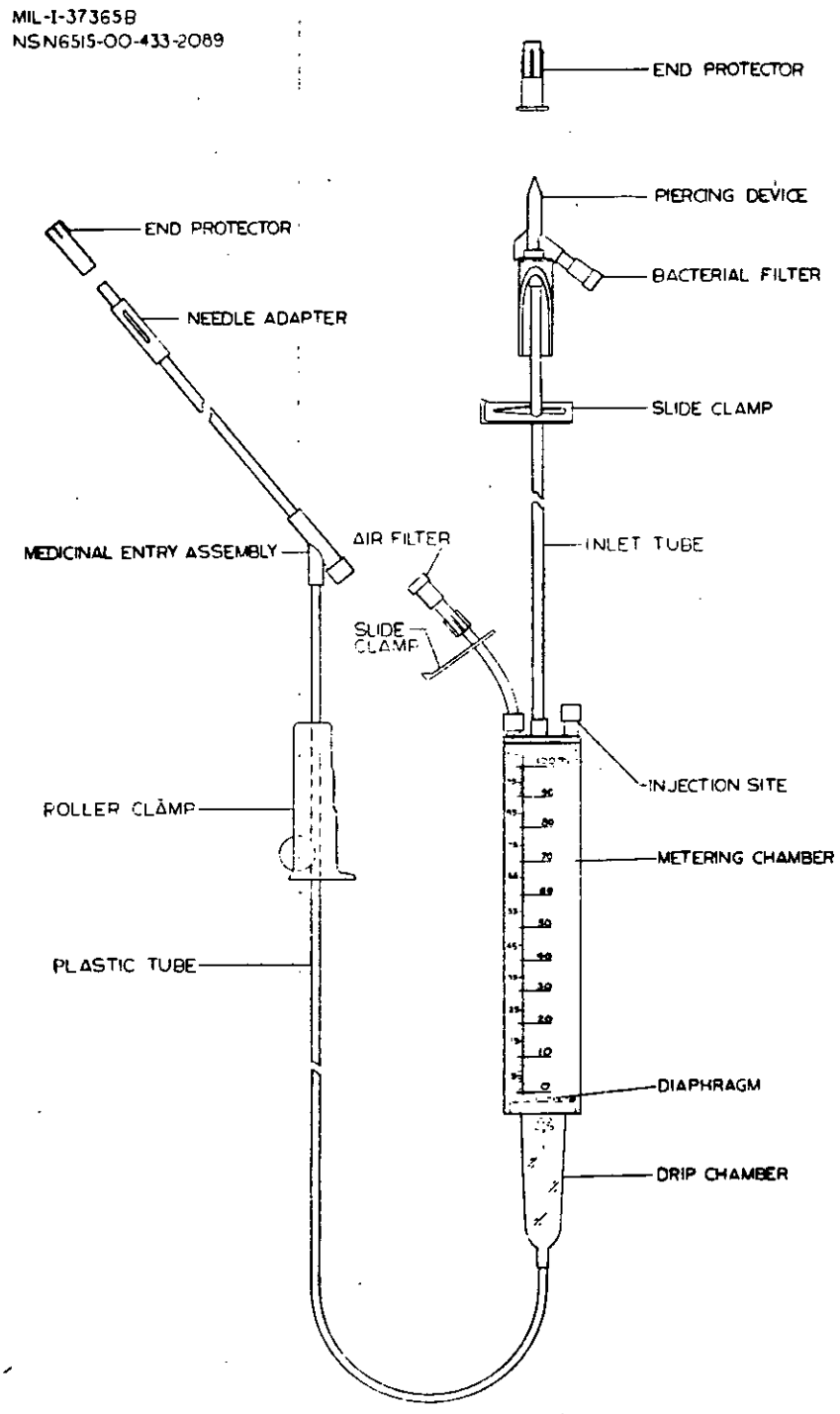


FIGURE 1. INTRAVENOUS INJECTION SET, WITH METERING CHAMBER, WITHOUT NEEDLE, DISPOSABLE

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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